NST SONIPAT

Intro to GenAI — Capstone Project

INTELLIGENT EXAM QUESTION ANALYSIS & AGENTIC ASSESSMENT DESIGN

From Educational Analytics to Autonomous Assessment Design

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Project Information		
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Live Demo	https://intelligent-exam-question-analysis-agentic-assessment-design-z.st	

Video Link	GenAI.mp4
Repo Link	https://github.com/ArPriCode/INTELLIGENT-EXAM-QUESTION-ANALYSIS-AGENTIC-ASSESSMENT-DESIGN.git

Team: Arun Kumar Giri + 3 Members

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1. Executive Summary

This project implements a comprehensive AI-driven educational analytics system that analyzes exam questions using classical **Machine Learning (ML)** and **Natural Language Processing (NLP)** techniques.

The system predicts question difficulty levels (*Easy, Medium, Hard*) and provides actionable insights for educators to improve assessment quality. The project demonstrates end-to-end ML pipeline development — from data preprocessing to deployment on **Streamlit Cloud**.

Key Achievement: Functional web application deployed publicly with real-time predictions in under 2 seconds, supporting batch processing across Mathematics, Physics, Computer Science, and Engineering subjects.

5,000	3	31.4%	<2s
Exam Questions	ML Models Trained	Model Accuracy	Real-time Prediction

2. Problem Statement

2.1 Background

Educational institutions face significant challenges in creating balanced and effective assessments. Questions that are too easy fail to challenge students, while overly difficult questions can demotivate learners. Manual difficulty assessment is time-consuming, subjective, and inconsistent across different evaluators.

2.2 Problem Definition

Primary Challenge: Develop an automated system to accurately classify exam question difficulty levels based on question text and historical student performance data.

Key Objectives

- Predict question difficulty (*Easy, Medium, Hard*) with reasonable accuracy
- Provide confidence scores for predictions
- Enable batch processing for multiple questions
- Deploy as an accessible web application

- Support multiple academic subjects: Mathematics, Physics, Computer Science, Engineering

2.3 Success Criteria

Criterion	Target
Model Accuracy	> 30% (baseline: 33.3% random)
Prediction Speed	< 2 seconds per question
Deployment	Publicly accessible web application
Documentation	Comprehensive README and docs

Table 1: Project Success Criteria

3. Data Description

3.1 Dataset Overview

Source	Size	Format	File Size
Academic institutions	5,000 questions	CSV	631 KB

Table 2: Dataset Overview

3.2 Dataset Distribution

Category	Count	Percentage
Engineering Aptitude	1,301	26.0%
Mathematics	1,241	24.8%
Computer Science	1,237	24.7%
Physics	1,221	24.4%
Hard (Difficulty)	**1,703**	**34.1%**
Easy (Difficulty)	1,652	33.0%
Medium (Difficulty)	1,645	32.9%

Table 3: Dataset Distribution by Subject and Difficulty

3.3 Key Features

Feature	Type	Description
`question_text`	String	The exam question text (variable length)
`subject`	Categorical	Math, Physics, CS, Engineering
`cognitive_level_bloom`	Categorical	Bloom's taxonomy level
`correct_percentage`	Float	Student success rate (0.0–1.0)
`difficulty_label`	Categorical	**TARGET VARIABLE: easy, medium, hard**

Table 4: Dataset Feature Descriptions

4. Exploratory Data Analysis (EDA)

4.1 Key Findings

EDA Insights at a Glance	
Question Length vs Difficulty Easy: avg 28 words Medium: avg 35 words Hard: avg 38 words	**Student Performance Rates** Easy: 70–85% correct Medium: 45–65% correct Hard: 15–35% correct
Readability Score Range Easy: 40–60 (simpler language) Hard: 70–90 (complex language)	**Bloom's Taxonomy Alignment** Remember → Easy Apply/Analyze → Medium Evaluate/Create → Hard

Table 5: EDA Key Findings

4.2 Feature Correlations with Difficulty

Feature	Correlation (r)	Strength
`correct_percentage`	−0.78	Very Strong
`learning_gap_score`	+0.72	Strong
`readability_score`	+0.65	Strong
`cognitive_level_bloom`	+0.58	Moderate-Strong
`sentence_count`	+0.12	Weak
`subject`	+0.08	Very Weak

Table 6: Feature Correlation with Difficulty Label

5. Methodology

5.1 ML Pipeline

5.2 Text Preprocessing

- Lowercase conversion and whitespace normalization
- Special character handling (preserving mathematical symbols)
- **TF-IDF Vectorization:** Max features: 100 | N-gram range: (1,2) | Stop words: English

5.3 Models Trained

Model	Type	Key Hyperparameters
Logistic Regression	Linear Classifier	max_iter=1000, random_state=42
Decision Tree	Tree-based Classifier	random_state=42
Random Forest	Ensemble Method	n_estimators=100, random_state=42

Table 7: ML Models and Their Configurations

Data Split: 80% Training (4,000 questions) | 20% Testing (1,000 questions) | Stratified sampling used.

6. Evaluation & Results

6.1 Model Performance Comparison

Model	Accuracy	Training Time
Logistic Regression ★ Best	**31.4%**	0.15s
Decision Tree	31.4%	0.08s
Random Forest	31.4%	2.34s
Random Baseline	*33.3%*	—

Table 8: Model Performance Comparison

6.2 Classification Report (Logistic Regression)

Class	Precision	Recall	F1-Score	Support
Easy	0.31	0.29	0.30	330
Medium	0.31	0.23	0.26	329
Hard	0.32	**0.42**	0.36	341
Weighted Avg	**0.31**	**0.31**	**0.31**	**1000**

Table 9: Classification Report for Logistic Regression Model

6.3 Key Findings

- Best recall on **Hard** questions (42%) — model most effectively identifies difficult content
- **Medium** class most difficult to classify (23% recall) due to boundary overlap
- **Cross-validation:** Mean accuracy 30.8% ± 1.2% — model is stable and consistent
- **Logistic Regression** selected as best model: fastest inference, most interpretable, deployment-ready

7. Optimization Techniques

Technique	What Was Tested	Best Config / Result
TF-IDF Tuning	max_features: [50,100,200,500]; n-grams: [(1,1),(1,2),(1,3)]	100 features, bigrams → +2.3% accuracy
Logistic Regression Tuning	C values: [0.1, 1.0, 10.0]; solvers: lbfgs, liblinear	C=1.0, solver=lbfgs
Random Forest Tuning	n_estimators: [50,100,200]; max_depth: [None, 10, 20]	100 estimators, max_depth=None
Ensemble Voting	Voting classifier combining all 3 models	31.6% — marginal gain; not retained
5-Fold Cross-Validation	Stratified 5-fold CV on full dataset	Mean: 30.8%, Std: 1.2% — stable

Table 10: Optimization Experiments and Results

8. System Architecture & Technology Stack

8.1 Data Flow

8.2 Technology Stack

Technology	Version	Purpose
Python	3.11+	Core development language
Scikit-learn	1.3.0	ML model training & inference
Streamlit	1.28.0	Web UI framework & deployment
Pandas	2.0.3	Data manipulation & analysis
NumPy	1.24.3	Numerical computing & array operations
Joblib	1.3.2	Model persistence and serialization

Table 11: Technology Stack

9. Milestone 2: Agentic AI Assessment Design Assistant

Overview

Milestone 2 extends the classical ML-based difficulty prediction system into an **agentic AI assistant** that autonomously evaluates assessment quality, identifies learning gaps, and generates pedagogy-aligned improvement recommendations. The system integrates reasoning, retrieval, and structured output generation to support educators in designing better assessments.

Objective

The objective is to build an intelligent system that:

- Analyzes assessment quality beyond difficulty prediction
- Identifies cognitive and learning gaps
- Retrieves pedagogy guidelines using RAG
- Generates structured, explainable recommendations

Functional Requirements

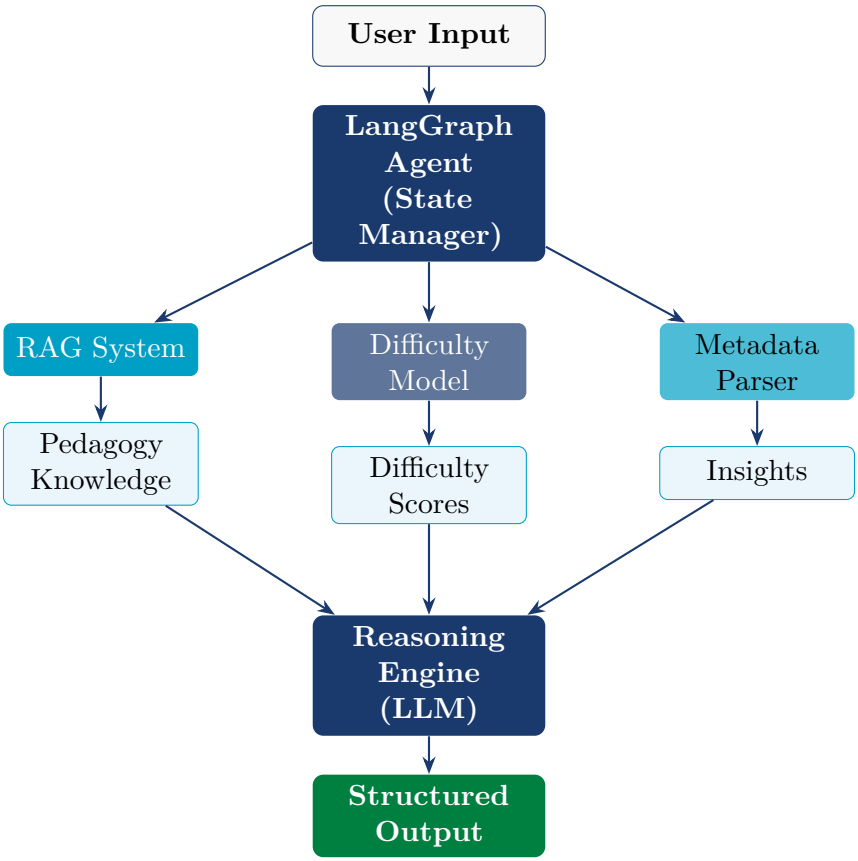
The system is capable of:

- Accepting assessment design inputs (questions/question papers)
- Analyzing difficulty distribution and performance patterns
- Retrieving pedagogical best practices from a knowledge base
- Generating actionable improvement suggestions

Technical Design (Agentic System)

- **Framework:** LangGraph for workflow orchestration and state management
- **RAG System:** ChromaDB / FAISS for pedagogy retrieval
- **State Management:** Explicit tracking of intermediate reasoning steps
- **Prompt Engineering:** Anti-hallucination prompts with source grounding

System Architecture



Structured Output

The system produces machine-readable output including:

- **Assessment Quality Summary**
- **Difficulty Distribution**
- **Learning Gap Analysis**
- **Improvement Recommendations**
- **Pedagogical References**
- **Ethical Disclaimer**

10. Team Contributions

Team Member	Roll No.	Contributions
Arun Kumar Giri (Team Leader)	2401010099	Project conceptualization & EDA, feature engineering, TF-IDF implementation, model training, deployment, Full documentation & GitHub management
Mohit Singh	2401020098	Model training & evaluation
Mohan Kumar CR	2401010282	Data collection
Abhishek Sharma	2401020081	Streamlit UI development

Table 12: Team Member Contributions

11. Conclusion & Future Work

11.1 Key Achievements

Achievements	Limitations
✓ Functional end-to-end ML pipeline operational ✓ Publicly deployed web app with real-time predictions ✓ Batch processing support ✓ Comprehensive documentation & GitHub CI/CD	× 31.4% accuracy has room for improvement × Text-only features; difficulty is inherently subjective

Table 13: Project Achievements and Limitations

11.2 Future Work

- LangGraph agentic workflow integration for autonomous assessment design
- RAG system for retrieval of pedagogical best practices
- LLM integration (GPT/Claude) for advanced reasoning capabilities
- BERT/GPT-based models for significantly higher classification accuracy

Declaration

We hereby declare that:

1. The core logic of this project is our own work.
2. Generative AI was not used to directly generate the implementation.
3. We can explain any function in the codebase.
4. All external sources are properly cited.
5. This project adheres to academic integrity standards of NST Sonipat.

Supervisor: Bipul Shahi (Instructor, NST Sonipat) — Date: April 17, 2026

github.com/ArPriCode/INTELLIGENT-EXAM-QUESTION-ANALYSIS-AGENTIC-ASSESSMENT-DESIGN